

SCENE 63: THE NHL ARSENAL — Master Treatment & Novel Therapies



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1. R-CHOP (Top Rack)

WHAT YOU SEE

Top weapon, R-CHOP label, indolent/aggressive tag

WHAT IT ENCODES

R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, prednisone) is the backbone for DLBCL and many aggressive B-cell NHL. 6 cycles standard. Cures ~60% of DLBCL. R = anti-CD20.

BOARD TRAP

Don't use R-CHOP for Burkitt — Burkitt needs intensive dose-adjusted regimens (DA-EPOCH-R or CODOX-M/IVAC) with CNS prophylaxis; R-CHOP undertreats it.

2. DA-EPOCH-R (Middle Rack)

WHAT YOU SEE

Middle weapon, DA-EPOCH-R label, 'double-hit' tag

WHAT IT ENCODES

Dose-adjusted EPOCH-R (etoposide added, infusional) is preferred for high-grade B-cell lymphoma with MYC + BCL2/BCL6 rearrangements (double/triple-hit) and primary mediastinal B-cell lymphoma. More intensive than R-CHOP.

BOARD TRAP

Double-hit lymphoma (MYC + BCL2) does poorly on R-CHOP — testing MYC/BCL2/BCL6 FISH is mandatory in DLBCL to escalate therapy.

3. R-CVP / Bendamustine (Indolent)

WHAT YOU SEE

Indolent-tagged option on rack

WHAT IT ENCODES

For indolent NHL (follicular, marginal zone) bendamustine-rituximab or R-CVP are gentler options; anthracycline often spared. Follicular lymphoma is incurable but indolent — treat when symptomatic (GELF criteria).

BOARD TRAP

Asymptomatic low-burden follicular lymphoma = watch-and-wait, not immediate chemo. Treating too early adds toxicity without survival benefit.

4. Rituximab Mechanism

WHAT YOU SEE

Antibody diagram, B-cell, ADCC/CDC/phagocytosis/apoptosis arrows

WHAT IT ENCODES

Rituximab is anti-CD20 monoclonal antibody. Kills B-cells via ADCC, complement-dependent cytotoxicity (CDC), phagocytosis, and direct apoptosis. Reactivates hepatitis B — screen HBsAg/anti-HBc before use.

BOARD TRAP

CD20 is the target — obinutuzumab is a glycoengineered anti-CD20 with stronger ADCC. CD20-negative relapse can occur, blunting rituximab.

5. CNS Prophylaxis

WHAT YOU SEE

Lighthouse 'CNS PROPHYLAXIS', IT MTX, HD-MTX, rules board

WHAT IT ENCODES

High CNS-relapse risk (high CNS-IPI, testicular, breast, renal/adrenal, paranasal, double-hit, HIV) warrants prophylaxis with high-dose IV methotrexate or intrathecal methotrexate. CNS relapse is rapidly fatal.

BOARD TRAP

Intrathecal methotrexate alone may be inadequate for parenchymal disease — high-dose IV MTX penetrates brain parenchyma better. Testicular DLBCL always needs CNS prophylaxis + scrotal RT.

6. CAR-T CD19 Bay

WHAT YOU SEE

CAR-T LAUNCH BAY, CD19 sign, soldier/cell, BAY 1/2/3 lines

WHAT IT ENCODES

CD19-directed CAR-T (axi-cel, tisa-cel, liso-cel) for relapsed/refractory DLBCL. Now 2nd-line for primary-refractory or early-relapse DLBCL (ZUMA-7, TRANSFORM beat autologous transplant). Living drug, single infusion.

BOARD TRAP

CD19 antigen loss is a mechanism of CAR-T relapse. Don't confuse line ordering — axi-cel/liso-cel moved to 2nd line; tisa-cel did not in TRANSCEND/BELINDA failed.

7. CAR-T Toxicity (CRS/ICANS)

WHAT YOU SEE

Launch bay soldier, bays implying monitoring

WHAT IT ENCODES

Cytokine release syndrome (fever, hypotension, hypoxia) treated with tocilizumab (anti-IL6R) ± steroids. ICANS (neurotoxicity) treated with steroids; tocilizumab doesn't cross BBB well. Grade and treat early.

BOARD TRAP

Tocilizumab is first-line for CRS but NOT for isolated ICANS — neurotoxicity needs corticosteroids. Mislabeling fever as infection delays CRS treatment.

8. Bispecifics (CD20xCD3)

WHAT YOU SEE

BIOTECH WING, BISPECIFICS, glofitamab/epcoritamab, CD20/CD3

WHAT IT ENCODES

Bispecific antibodies (glofitamab, epcoritamab, mosunetuzumab) engage CD3 on T-cells and CD20 on tumor — off-the-shelf T-cell redirection. Used in R/R DLBCL and follicular lymphoma. Also cause CRS but milder, stepped dosing.

BOARD TRAP

Bispecifics target a DIFFERENT antigen (CD20) than CD19 CAR-T — so CD19-loss CAR-T relapse can still respond to CD20 bispecifics.

9. Polatuzumab (ADC)

WHAT YOU SEE

ADC label, CD79b target arrow

WHAT IT ENCODES

Polatuzumab vedotin is an anti-CD79b antibody-drug conjugate (MMAE). Pola-R-CHP replaced vincristine with pola in DLBCL (POLARIX) improving PFS. Also in R/R DLBCL with bendamustine-rituximab.

BOARD TRAP

Target is CD79b, not CD19/CD20 — distinct from CAR-T and rituximab. POLARIX showed PFS but not clear OS benefit, so not universally adopted first-line.

10. Brentuximab (CD30)

WHAT YOU SEE

BRENT label, CD30 differentiation arrow, biotech wing

WHAT IT ENCODES

Brentuximab vedotin is anti-CD30 ADC (MMAE) — used in Hodgkin lymphoma (A+AVD) and CD30+ peripheral T-cell lymphoma (ECHELON-2 with CHP). Peripheral neuropathy is dose-limiting.

BOARD TRAP

Target is CD30 — works only on CD30-expressing tumors (Hodgkin RS cells, ALCL). Replaces bleomycin in A+AVD to avoid pulmonary toxicity.

11. Tafasitamab + Lenalidomide

WHAT YOU SEE

Implied CD19-directed line in launch bay

WHAT IT ENCODES

Tafasitamab is an Fc-enhanced anti-CD19 monoclonal antibody; with lenalidomide (L-MIND) it treats transplant-ineligible R/R DLBCL. Provides CD19 targeting without CAR-T logistics.

BOARD TRAP

Because tafasitamab targets CD19, prior tafasitamab may select CD19-low clones that compromise later CD19 CAR-T efficacy — sequencing matters.

12. Hepatitis B Reactivation

WHAT YOU SEE

Rituximab mechanism panel implying B-cell depletion risk

WHAT IT ENCODES

Any anti-CD20 (rituximab/obinutuzumab) profoundly depletes B-cells and reactivates HBV — potentially fatal fulminant hepatitis. Screen HBsAg and anti-HBc; give entecavir/tenofovir prophylaxis if positive.

BOARD TRAP

Anti-HBc positive with negative HBsAg (occult/resolved HBV) STILL needs antiviral prophylaxis with rituximab — checking only HBsAg misses these.
